



选取杆2位于垂直最低点, ($\theta_2=0$) 处为零势能点.

$$q = [d_1, \theta_2]$$

$$K_1 = \frac{1}{2} m_1 \dot{d}_1^2$$

$$K_2 = \frac{1}{2} m_2 (\dot{d}_1 + l_2 \dot{\theta}_2 \cos \theta_2)^2 + \frac{1}{2} m_2 l_2^2 \dot{\theta}_2^2 \sin^2 \theta_2 + \frac{1}{2} I_2 \dot{\theta}_2^2$$

$$= \frac{1}{2} m_2 \dot{d}_1^2 + m_2 l_2 \dot{\theta}_2 \cos \theta_2 \dot{d}_1 + \frac{1}{2} m_2 l_2^2 \dot{\theta}_2^2 + \frac{1}{2} I_2 \dot{\theta}_2^2$$

$$\text{总动能 } K = K_1 + K_2$$

$$\text{杆1势能: } U_1 = m_1 g l_2$$

$$\text{杆2势能: } U_2 = m_2 g l_2 - m_2 g l_2 \cos \theta_2$$

$$\text{总势能: } U = U_1 + U_2$$

$$\text{偏导数: } \frac{\partial K}{\partial q} = \begin{bmatrix} 0 \\ -m_2 l_2 \dot{\theta}_2 \sin \theta_2 \dot{d}_1 \end{bmatrix}$$

$$\frac{\partial K}{\partial \dot{q}} = \begin{bmatrix} m_1 \dot{d}_1 + m_2 (d_1 + l_2 \dot{\theta}_2 \cos \theta_2) \\ m_2 l_2 \cos \theta_2 \dot{d}_1 + m_2 l_2^2 \dot{\theta}_2 + I_2 \dot{\theta}_2 \end{bmatrix}$$

$$\frac{\partial U}{\partial q} = \begin{bmatrix} 0 \\ m_2 g l_2 \sin \theta_2 \end{bmatrix} \quad \frac{\partial U}{\partial \dot{q}} = 0$$

$$\text{动力学标准型: } \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = \tau$$

$$\Rightarrow \begin{bmatrix} m_1 + m_2 & m_2 l_2 \cos \theta_2 \\ m_2 l_2 \cos \theta_2 & m_2 l_2^2 + I_2 \end{bmatrix} \ddot{q} + \begin{bmatrix} -m_2 l_2 \sin \theta_2 \dot{\theta}_2^2 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ m_2 g l_2 \sin \theta_2 \end{bmatrix} = \tau$$